NTD & BART AFC EMPIRICAL STUDY

Longitudinal System Divergence & Spatial Trip Dynamics

A comprehensive analysis of post-2020 public transit demand across the San Francisco Bay Area. Examining NTD time-series data (2018–2025) and BART automated fare collection (AFC) origin-destination matrices to evaluate modal decoupling, trip length growth, and financial sustainability.

Scope: BART vs SFMTA vs AC Transit

Data Window: 2018 – 2025

BART RECOVERY PLATEAU — Stalled

45.0%

55.6M annual UPT in 2025 vs 125M+ pre-pandemic baseline.

Peak hybrid work impact

AC TRANSIT & SFMTA — Resilient

72% – 76%

Local bus networks recovered essential & non-commute trips faster.

Local intra-urban mobility

AVG TRIP LENGTH SHIFT — +6.82%

14.88 mi

Grew from 13.93 mi (Oct 2019) to 14.88 mi (Oct 2024) per passenger.

Short transbay commutes fell

UNIT COST INFLATION — +166.7%

$16.00

Operating cost per boarding rose from ~$6.00 (FY19) to ~$16.00 (FY25).

Farebox ratio: 67% → 24%

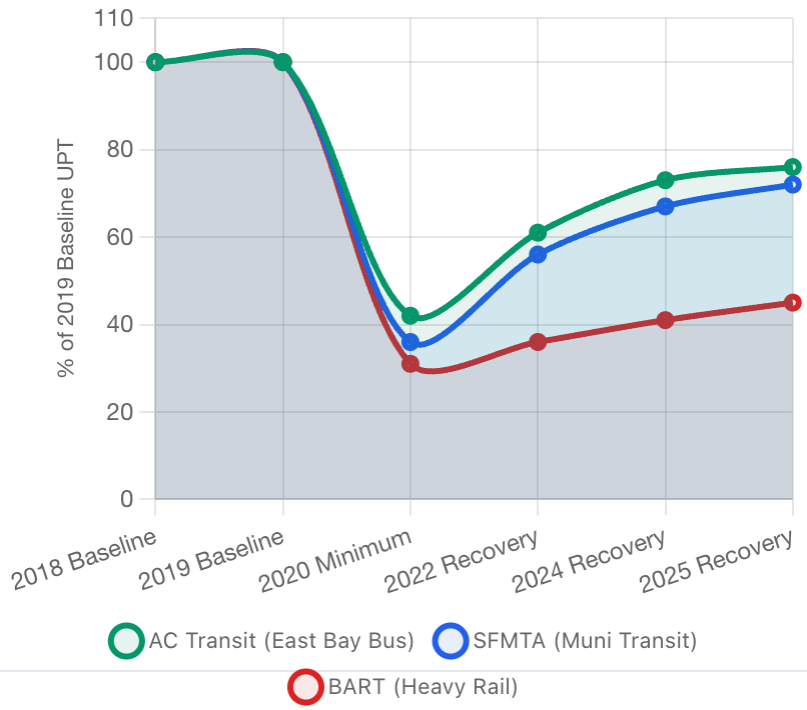
SECTION 1

System-Wide Recovery Divergence Curves (2018–2025)

Post-2020 transit recovery trajectories across the Bay Area highlight a structural decoupling between high-capacity heavy rail (BART) and localized urban/bus networks (SFMTA and AC Transit). High remote work adoption in knowledge sectors severely reduced weekday peak commuting into downtown San Francisco and Oakland. By early 2025, hybrid arrangements reached 63% and fully remote stood at 11%, anchoring BART's passenger volume at 45% of baseline, whereas neighborhood and essential service trips buoyed bus networks up to 72%–76% recovery.

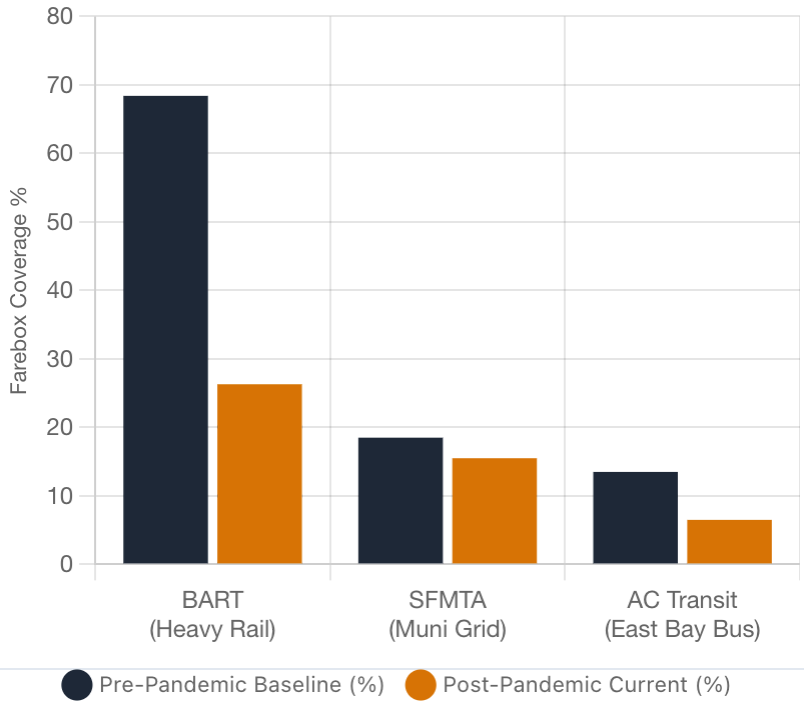
Agency Ridership Recovery (% of Pre-Pandemic Baseline)

NTD Monthly Module Adjusted Unlinked Passenger Trips (UPT)



Pre vs. Post Pandemic Farebox Cost Coverage

Operating expenses recovered directly through passenger fares



Key Modal Insight

While 76% of pre-pandemic patrons had returned to BART by late 2024, total boardings remained at only 45% because individual commuters travel fewer days per week (2.5 to 3 days on average). In contrast, SFMTA and AC Transit benefit from short-distance multidirectional trips (school, medical, shopping) where midday and weekend ridership stabilized at 85%–95% of baseline.

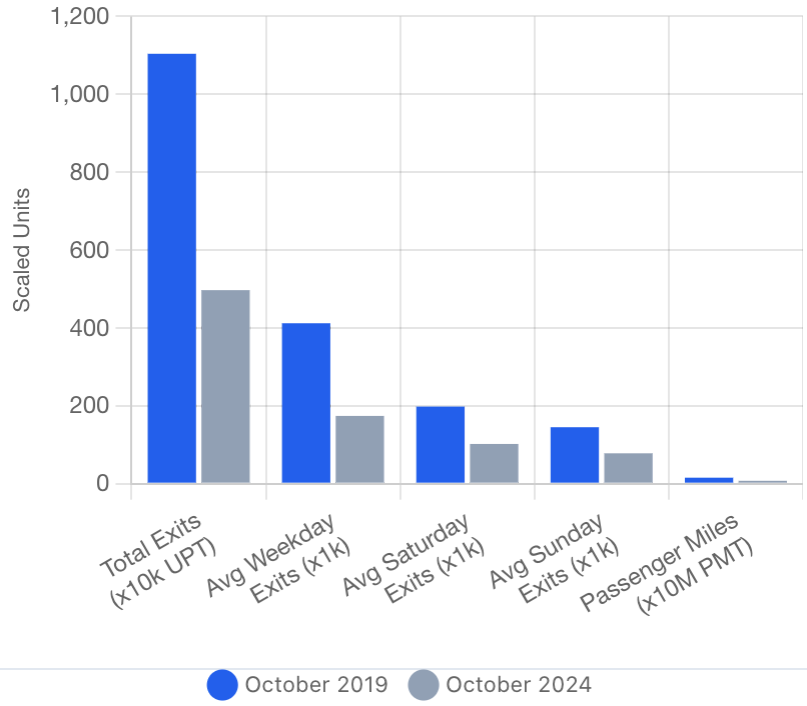
📍 SECTION 2

BART Origin-Destination Matrix & Trip Distance Dynamics

Analysis of BART's closed automated fare collection (AFC) system comparing October 2019 and October 2024 uncovers a distinct spatial restructuring. Even though total monthly exit volumes fell by 55.01%, the volume-weighted average passenger trip length ($\bar{D}$) expanded from 13.93 miles to 14.88 miles (+6.82%). This shift stems from an asymmetrical loss in short transbay commutes (6–11 miles) alongside the addition of long suburban origin-destination pairings following the June 2020 Silicon Valley Extension Phase 1 opening (Milpitas and Berryessa).

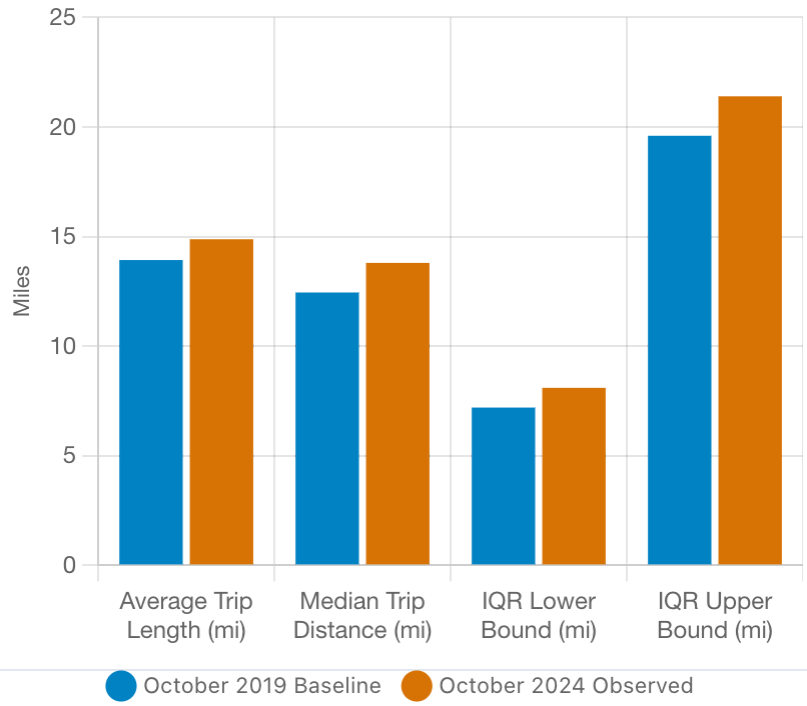
System Volume Contraction (Oct 2019 vs Oct 2024)

Comparison of exit counts and passenger miles traveled



Trip Length & Distance Distribution Metrics

Average, median, and interquartile range (IQR) distance in miles



🔻 Short Transbay Decline

Station pairs connecting East Bay core (Rockridge, MacArthur, 19th St) to downtown SF fell >60%, as tech sector commuters transitioned to hybrid schedules.

🚆 Outer Suburban Resilience

Longer distance commutes from outer termini (Antioch, Dublin, Warm Springs, Millbrae) experienced a milder 45%–50% reduction.

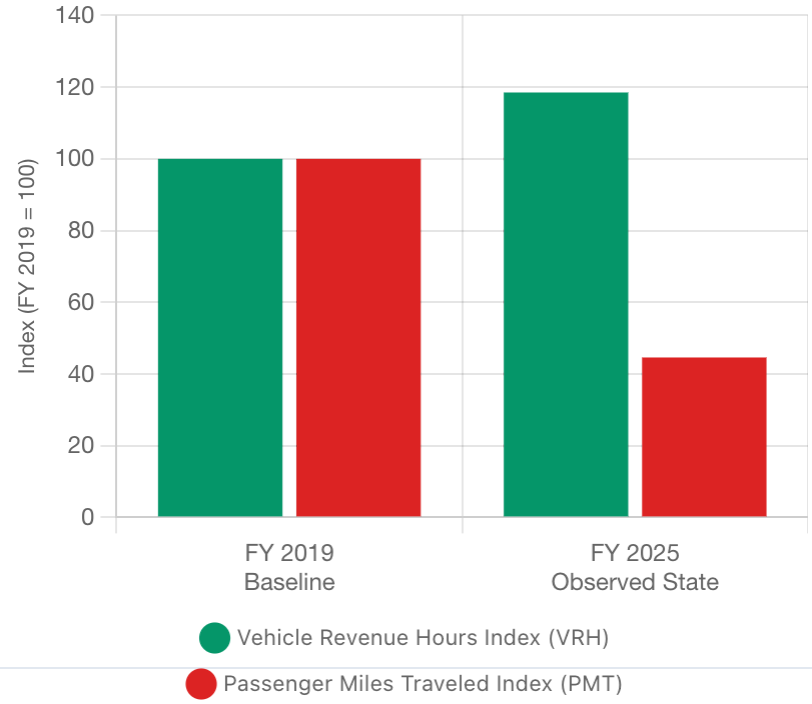
✈️ Airport Trip Recovery

Ridership to SFO and OAK international airports recovered faster than central office trips, reinforcing long-haul passenger miles.

⚡ SECTION 3

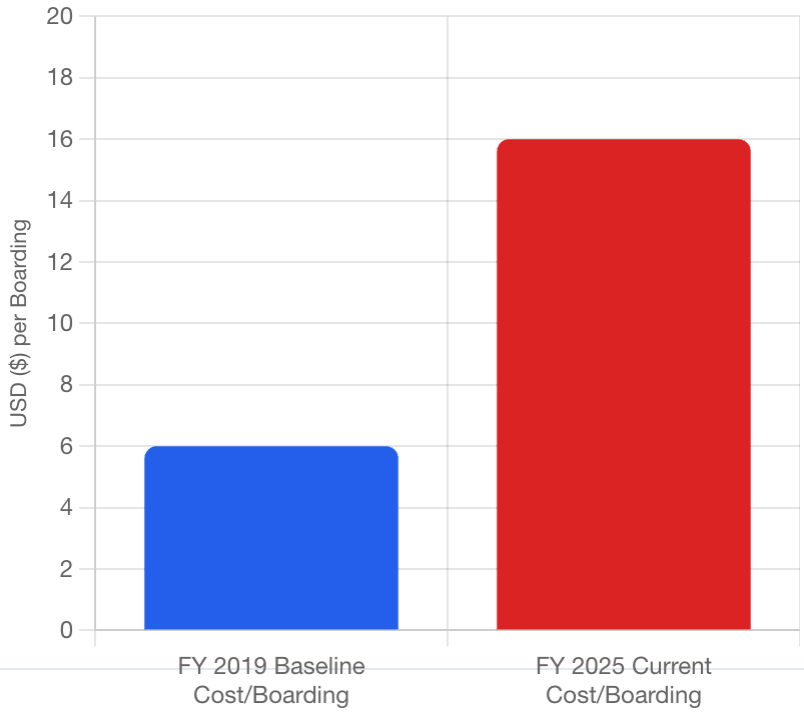
Capacity Utilization, Service Decoupling & Unit Costs

Heavy rail operations entail high fixed capital and maintenance costs that cannot scale down proportionately with ridership drops. In FY 2025, BART provided 18.5% more vehicle revenue hours (VRH) than in FY 2019 to improve service frequency and safety. However, passenger miles traveled (PMT) hovered at only 44.6% of pre-pandemic levels. This 73.9 percentage point gap between service supplied and demand captured drove operating cost per boarding up from $6.00 to $16.00.



Operating Cost Per Passenger Boarding ($ USD)

Pre-pandemic efficiency vs. post-pandemic cost inflation



SECTION 4

Data Architecture & Dataset Access Repositories

The analytical findings in this report are anchored in official repositories provided by the Federal Transit Administration (FTA) National Transit Database and BART's Automated Fare Collection system. Direct download mirrors and documentation endpoints are detailed below.

Data System / Portal	Format	Primary Attributes	Access Endpoint Link
FTA NTD Monthly Module (Adjusted)	`XLSX / API`	Agency ID, Mode, Service Type, UPT, VRM, VRH, VOMS	FTA NTD Portal 🔗
FTA DOT Open Data Platform API	`JSON / Socrata`	National transit monthly time series, API endpoints	DOT Open Data API 🔗
BART AFC Hourly O-D Directory	`CSV.GZ Archive`	Hourly origin, destination, entry/exit gates count	BART AFC Directory 🔗

Data System / Portal	Format	Primary Attributes	Access Endpoint Link
BART Monthly Reports Hub	ZIP / XLSX	Station matrix aggregated by Weekday, Sat, Sun	BART Reports Landing 🔗
BART Oct 2024 Performance Snapshot	PDF Report	Monthly exits summary, peak factors, system load	Oct 2024 Snapshot PDF 🔗

🚀 SECTION 5

Policy Implications & Transit Realignment Strategy

The structural shift from peak commuter reliance to hybrid travel patterns necessitates operational, financial, and policy updates across Bay Area transit governance.

⏱️ All-Day Clock-Face Scheduling

Service Restructuring

Transition schedules away from morning/evening peak concentrations toward uniform 15-minute clock-face headways spanning midday, evening, and weekend windows to capture non-commute, leisure, and regional personal travel.

💳 Regional Fare Integration

Clipper Transfer Subsidies

Expand multi-agency Clipper fare integration to eliminate double-fare penalties for local bus-to-rail transfers, encouraging short-distance trips on heavy rail trunk lines.

🏛️ Permanent Subsidy Mechanisms

Dedicated Revenue Sources

Replace expiring federal COVID relief funds with dedicated regional tax measures (e.g., sales, property assessments, congestion pricing) as farebox recovery drops permanently below 30%.

Transit-Oriented Development

Station Area Density

Accelerate high-density residential housing development on BART station parking lots (e.g., Millbrae, Walnut Creek, Berryessa) to cultivate localized walk-up ridership.

Bay Area Public Transit Longitudinal Study & Matrix Analytics

Data Sources: Federal Transit Administration NTD & BART AFC Origin-Destination Database (2018–2025)